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Biofilm formation and antifungal resistance in
candida species: A comparative study of Albicans
and Non albicans strains in hematology ICU

**A Thesis submitted for Fulfillment of
MSC degree in Medical Microbiology and Immunology
By**

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Introduction

Candida genus is considered a part of human microbiota [1]. However, in certain pathogenic conditions, it can be a leading cause of systemic invasive infections. *Candida spp* are the fourth cause of nosocomial blood stream infections and had a 37% mortality rate within 30-day duration[2, 3].

Early laboratory diagnosis of *Candida* invasive infections reduces the mortality rate from 40% to 15% after the start of appropriate antifungal therapy [4]

There are several *Candida* species such as *Candida albicans* , *Candida glabrata*, *Candida tropicalis*, *Candida auris* but *C. albicans* remains the most frequent species isolated [5].

Despite the fact that *C. albicans* is still regarded as a significant pathogen of candidal infection , an ongoing shift from *C. albicans* to *non-albicans Candida (NAC) spp*. Was reported by several countries. *C. albicans*, *C. tropicalis*, *C. parapsilosis*, *C. krusei* and *C. glabrata* are responsible for over 90% of cases of candidal infection [6]

Several pathogenic virulence factors are encoded by *C. albicans* genes and assist the fungus to invade the host tissues leading to infections as the capacity of *C. albicans* to change from the budding yeast form to filamentous form, expression of adhesion factors, and secretion of hydrolytic enzymes [7]

Many microbes, including yeasts, form biofilms as one of the major virulence factors. A biofilm is a community of microbes attached to biotic or abiotic surfaces embedded within an extracellular matrix (ECM) [8]

Candida infections often get therapeutic failure, mostly as a result of antifungal resistance that is caused by several mechanisms including biofilm production as biofilm-producing strains show significant increased resistance to antifungal drugs and host immunity[9] .

More unusual species have been known to induce candidemia, which poses a new risk to hospitalized patients' health. *Candida auris* is an emergent pathogen that was first described in 2009 in Japan [10].

C. auris can be challenging to identify in the laboratory using conventional methods and its detection may therefore be underestimated

Importantly, *C. auris* isolates are resistant to fluconazole and frequently show multidrug resistance[11]

C. auris has a high capacity for dissemination via contaminated surfaces or horizontal patient-to-patient transfer, which is unusual in other *Candida* species[12]

Due to the serious threat posed by *C. auris*, the CDC issued an alert to healthcare facilities to notify each case of *C. auris* infection to local or public health authorities[12]

Aim of the work

- Describe the distribution of different candida species including candida auris as the causative agent in hematology ICU infection
- To determine the susceptibility pattern of our local isolated *Candida* strains which is essential for optimal management of fungal infection.
- Detection of biofilm formation by conventional and molecular methods
- Comparison between *C. albicans* and non-albicans in the prevalence of biofilm formation and biofilm -forming genes
- Find the association between antifungal resistance and biofilm formation in candida strains isolated from patients
- Determination of clinical factors associated with occurrence of infections.

Materials and methods :

Study subjects: patients admitted to hematology intensive care

Inclusion criteria:

Eligible participants in the study were those with invasive infection and had one or more of the following criteria:

- Suppressed general body immunity.
- Suppressed local body immunity as a result of trauma or invasive procedures.

- Resistant to antibiotic therapy.
- Negative samples for bacterial examination and culture.
- Critical life threatening infections.

Exclusion criteria:

- Patients who received antifungal therapy within 3 days prior to sample collection.
- Resistant to antibiotic therapy.

1 - Samples : Samples were collected from the patients according to their clinical presentation and different localizing signs and symptoms.

These samples included:

- Sputum samples or endotracheal tube aspirates in patients with respiratory tract infections.
- Urine samples in patients with urinary tract infections.
- Blood cultures in patients with suspected fungaemia.

Sample size:

- The sample size was calculated using the StatCalc feature in EPI-Info version 5, based on a prevalence rate of 70% according to (Wilson and Lorenz, 2023) and a permissible margin of error (d) of no more than 0.05, using 95% confidence interval and 80% power of the test. This calculation indicated that a total sample size of 323 was necessary.

2 - isolation of candida:

Culture:

1- **sabouraud dextrose agar (SDA)** : After 24-48hr incubation at 37°C, morphological identification are white to cream coloured, smooth, glabrous and yeast-like in appearance

2- **Chromogenic Candida Agar (CCA)**: Chromogenic Candida agar was used for differentiation of Candida species according to coloration and colony morphology [13]

- C. albicans appear as light green coloured smooth colonies, C. tropicalis appear as blue to metallic blue coloured raised colonies. C. glabrata colonies appear as cream to white smooth colonies, while C.krusei appear as purple fuzzy colonies.
- Nutrient broth supplemented with 20% glycerol was used to preserve confirmed Candida isolates at – 80 °C

3 - Identification of candida : isolated colonies are further identified by:

Film stained by Gram stain showing violet budding yeast cell.

Urease test :

Purpose: Differentiates *Candida albicans* from *Candida krusei* and *Candida tropicalis*

Principle: *Candida* species that produce urease enzyme hydrolyze urea, forming ammonia, which increases pH and changes indicator color.

result :

- **Positive Reaction** : Development of an intense magenta to bright pink color .
- **Negative Reaction** : No color change

Germ tube formation test: Positive test is when a short hyphal (filamentous) extension arising laterally from a yeast cell, with no constriction at the point of origin[14]

Sugar Fermentation Test [15] :

Purpose: -Determine the ability of *Candida* species to acidify carbohydrate substrate in the medium detected by a pH indicator (phenol red).

Method :

- 1-The purple broth was prepared by mixing peptone broth with a pH indicator.
- 2-Each tube of purple broth was inoculated with specific carbohydrates (glucose, lactose, sucrose, and mannitol) and isolated colonies from a 24-hour pure culture.
- 3- A Durham tube was provided in tubed broth media to collect the gas produced during fermentation.
- 4- The pH indicator changed from purple to yellow when the acid produced from the fermentation was more than the alkaline.

4 - Detection of biofilm formation :

- Congo red agar method : Candida isolates produce black colonies on Congo red agar during a 48-hr Incubation at 37 °C, were considered biofilm forming isolate .
- tissue culture plate (TCP) method : a technique used to assess biofilm formation by candida . It involves growing cells in microtiter plates and then quantifying the amount of biofilm that forms on the wells by recording The optical density (OD) at 595nm and interpreting the result[16]

5-Antifungal Susceptibility test Various antifungal discs from HiMedia were utilized, categorized as follows:

A. Azoles

- Fluconazole (10 µg)
- Ketoconazole (10 µg)
- Voriconazole (1 µg)
- Clotrimazole (10 µg)
- Miconazole (50 µg)

B. Polyenes

- Nystatin (50 µg)
- Amphotericin B (20 µg)
- C. Echinocandins
- Micafungin (10 µg)
- Caspofungin (25 µg)

6-Molecular identification of genes of biofilm formation by conventional PCR[17][18] :

Target Gene	Primer Sequence (5'-3')	PCR conditions	Size (bp)	Reference
HWPI	F: CCATGTGATGATTACCCACA	Initial denaturation: 1 cycle of 94 °C for 4 min, Denaturation: 35 cycles of 94 °C for 30 s, Annealing: 52 °C for 1 min, Extension: 72 °C for 2 min. A final extension cycle was performed at 72 °C for 5 min	572	Inci et al.
	R: GCTGGAACAGAAGATTCAGG			
ALS1	F: CCATCACTGAAGATATCACCACA		318	
	R: TGGAGCTTCTGTAGGACTGGTT			
SAP 2	F: AACAAACAACCCACTAGACATCACC		178	Lima et al.
	R: TGACCATTAGTAACTGGGAATGCTTTAGGA			

7-Molecular identification of Candida auris by PCR amplification of species specific GPI protein-encoding genes[12]:

*

Systematic ORF name	Primer name	Primer sequence (5'- > 3')	Amplicon size (bp)
QG37_03410	03410_F	GCCGCTAGATTGATCACCGT	137
	03410_R	TAGGTGTGGGTACCCTTGGT	

- identification and Speciation of Candida auris Isolates is confirmed by PCR following by sequencing using ITS1F/ITS4R primers for the variable internal transcribed spacers ITS1 and ITS2 regions, located between universally conserved genes 18S, 5.8S, and 28S and NL1F/NL2R primers used to detect the D1–D2 region located at the 50 end of the gene 28S encoding for the large nuclear ribosomal subunit[19]

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Thank You

